ADVANCED MATERIALS INTERFACES
Open Access
www.advmatinterfaces.de

Check for updates

Annealing Temperature Effect on the Physical Properties of NiO Thin Films Grown by DC Magnetron Sputtering

*Sergei Timoshnev, Alexey Kazakin, Ksenia Shubina, Valentina Andreeva, Elizaveta Fedorenko, Aleksandra Koroleva, Evgeniy Zhizhin, Olga Koval, Alina Kurinnaya, Alexander Shalin, Vjaceslavs Bobrovs, and Yakov Enns**

Nickel oxide is a promising material for transparent electronics applications. This semiconductor demonstrates the possibility of modifying its physical properties depending on the method of growth and subsequent processing. Here the effects of the discharge power are reported during reactive dc magnetron sputtering, as well as the modes of subsequent annealing of NiO films, on their structural, electrical, and optical properties. NiO films are annealed at various temperatures both in an oxygen-containing environment and under vacuum conditions. Deposited NiO films have a polycrystalline structure with a preferred orientation (200) for the low discharge power mode and (111) for the high discharge power mode. However, obtained NiO films exhibit crystallinity improvement after annealing. The presence of both Ni^{2+} and Ni^{3+} oxidation states in the deposited films is found. In addition, it is shown that the relative carrier concentration (Ni^{3+}/Ni^{2+} peak area ratio) can be controlled by choosing the NiO film preparation mode. The trend in this ratio corresponds to the trend in film conductivity and the number of free-charge carriers. The deposited films are semitransparent, and the estimated optical bandgap of NiO is in the range from 3.50 to 3.74 eV.

1. Introduction

In recent years, thin film technology based on oxide materials has a high potential for designing in semiconductor devices. Transparent conductive oxide materials have attracted a lot of attention due to their controlled electrical conductivity and optical transmission depending on growth and post-processing modes. Nickel oxide (NiO) has received much attention as a highly promising semiconductor material with a wide potential for both power applications (rectifying diode structures[1]) and optoelectronic applications (solar cells[2–5]; ultraviolet sensors[6–8]). Nickel oxide is a direct-gap semiconductor with a large bandgap of 3.6–4.0 eV,[9] which has excellent thermal and chemical stability and high transparency.[10] It exhibits natural hole conductivity, as a rule, due to a large number of intrinsic defects – such as mainly Ni (V_{Ni}) vacancies and O (V_O) vacancies.[11] However, although V_{Ni} are considered to be efficient acceptors in NiO, they also degrade the transparency of the material in the visible region. It has been found that V_{Ni} present in NiO grown in a high O environment results in significant subband absorption compared to V_O in NiO grown in a high Ni environment.[11] The publication explains this behavior by the high concentration of gap states in NiO:V_{Ni}, which resembles a semimetallic system, compared to the shallower subgap states in NiO:V_O. Therefore, for applications of

S. Timoshnev, A. Kazakin, K. Shubina, O. Koval, Y. Enns
Alferov University
Saint-Petersburg 194021, Russia
E-mail: enns_jakov@spbau.ru

S. Timoshnev
Ioffe Institute
Saint-Petersburg 194021, Russia

V. Andreeva, E. Fedorenko
Peter the Great St.Petersburg Polytechnic University
Saint-Petersburg 195251, Russia

A. Koroleva, E. Zhizhin
Research Park
Saint Petersburg State University
Saint Petersburg 199034, Russia

A. Kurinnaya, A. Shalin
Center for Photonics and 2D Materials
Moscow Institute of Physics and Technology
Dolgoprudny 141700, Russia

A. Shalin
Faculty of Physics
Moscow State University
Moscow 119991, Russia

V. Bobrovs
Institute of Telecommunications
Riga Technical University
Riga 1048, Latvia

The ORCID identification number(s) for the author(s) of this article can be found under https://doi.org/10.1002/admi.202300815

© 2024 The Authors. Advanced Materials Interfaces published by Wiley-VCH GmbH. This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

DOI: 10.1002/admi.202300815

transparent optoelectronic devices, it is necessary to study technological methods for optimizing intrinsic defects in NiO to simultaneously increase the conductivity and transparency of films.

NiO films can be obtained by various methods, including electron beam evaporation,[12] magnetron sputtering,[13–15] atomic layer deposition,[16] pulsed plasma deposition,[17] thermal oxidation,[18] centrifugation,[19] chemical vapor phase deposition,[20] etc. Among these methods, magnetron sputtering is widely used due to the high quality of the film, selective deposition over the area, low cost, and high deposition rate. Films are deposited using both rf magnetrons and high-purity Ni targets or stoichiometric NiO[21] and dc magnetrons with Ni targets[22] or nickel alloys.[23] In both cases, deposition occurs with the addition of O_2 to the working gas mixture. In this case, a high concentration of acceptors can be easily achieved due to spontaneously formed Ni vacancies or interstitial O atoms,[24] which is provided by the choice of reactive sputtering or additional thermal annealing.

A decrease in the oxygen content in the working gas mixture during magnetron sputtering of a film from a NiO target is accompanied by an improvement in the quality of the film.[25] Thus, the deposited NiO film has a crystal structure and orientation (111), but with an increase in the oxygen content in the mixture of working gases, a decrease in the crystallinity of the film, an increase in charge carriers concentration, and thus a decrease in their mobility are observed. Subsequent high-temperature annealing in an oxygen atmosphere leads to a decrease in the concentration of intrinsic defects inside the film and an increase in the width of the optical bandgap.[21] A similar trend is also observed with an increase in the sputter power.[13]

The conditions for thermal annealing of the film are also of high importance. As a rule, to recrystallize the film and reduce point defect density, annealing is used in an oxygen-containing environment at temperatures of 200–500 °C and a duration of 10 to 60 min.[21,22,25] However, it is argued that atmospheric annealing does not allow the removal of the NiOOH phase on the surface of an oxide semiconductor.[25] It is assumed that there is a high probability that the NiOOH phase is formed after heat treatment by reaction with moisture from the atmosphere. The use of vacuum annealing can promote thermal decomposition of the NiOOH phase. In addition, since the NiO phase is more stable than the Ni_2O_3 phase at low oxygen partial pressure,[26] vacuum heat treatment should reduce the content of this phase on the surface. However, the absence of a sufficient number of studies of the applied annealing under vacuum conditions does not allow a sufficient assessment.

In this work, we studied in detail the effect of the discharge power during dc magnetron sputtering of NiO films and the modes of their subsequent annealing on the structural, electrical, and optical properties.

2. Experimental Section

2.1. NiO Films Formation

NiO was deposited to a previously cleaned surface of commercial Corning 0211 glass. The glass was cleaned by boiling in acetone for 10 min at a temperature of ≈56 °C, followed by washing in isopropyl alcohol and water and drying in a stream of pure nitrogen.

NiO films were formed using dc magnetron sputtering from a Ni target (a disk 106 mm in diameter and 1.3 mm thick made of H95 grade nickel (99.95% pure)) in a working gas mixture of 25% O_2/75% Ar at a operating pressure of 3 mTorr. The deposition was carried out at room temperature. The distance from the target to the substrate holder with the samples was 20 cm. Before deposition, the vacuum chamber was evacuated to a residual pressure of 3×10^{-6} Torr, after which the ion cleaning of the sample surface was carried out in pure argon plasma at a current of 30 mA for 5 min. The ion cleaning power was 45 W. Immediately before deposition on the wafers, the target was cleaned of oxides and other possible contaminants by sputtering it in an Ar environment for 10 min. In this way, the same initial conditions were achieved for all deposition modes. The deposition of nickel oxide was carried out in two modes – SP1 and SP2, which differed qualitatively in the nature of the sputtering of the target material.

The SP1 mode consisted of deposition at a low discharge power of 100 W (0.3 A, 330 V), in which the target surface remained completely passivated with oxygen during the entire sputtering process, and the deposition rate was 1.6 nm min^{-1}. The target in the SP1 mode was preliminarily sputtered onto an empty shutter for 10 min, after which the samples were deposited for 90 min.

The SP2 mode consisted in sputtering the unpassivated target at a high discharge power of 265 W (0.5 A, 530 V), as a result of which the deposition rate was 36 nm min^{-1}. The increased discharge power contributed to the absence of oxide passivation of the target, which, in addition to increasing the deposition rate and energy for the crystallite formation, also significantly changes the nature of the deposition. The target was preliminarily sputtered in pure argon for 10 min to stabilize the discharge, after which it was deposited on the samples for 5 min.

2.2. Methods for Studying NiO Films

Film thicknesses were determined using an Ambios XP-1 contact profilometer along a stepped profile formed during deposition.

After deposition, the NiO films were subjected to thermal annealing in the atmospheric environment (oxygen-containing) or vacuum. In the case of heat treatment in an atmospheric environment, annealing was carried out at temperatures of 200–550 °C with a duration of 5–120 min. To minimize the diffusion of glass components into the NiO film layer, the choice of the maximum annealing temperature was determined by the parameters of the glass substrates. The beginning of the process was the placement of the sample in a heated furnace. At the end of the annealing process, the sample was taken out and cooled at room temperature.

Vacuum annealing was carried out in a specialized vacuum chamber. After placing the sample, the chamber was preliminarily degassed. The pressure in the chamber was 3×10^{-5} Torr. The annealing temperature was 200–400 °C with reaching the operating temperature within 15–30 min. Temperature control was carried out with a Pt-1000 thermometer placed directly on the sample using a pressure contact. After reaching the working temperature, the sample was kept for 15 min and cooled in a vacuum chamber to room temperature.

© 2024 The Authors. Advanced Materials Interfaces published by Wiley-VCH GmbH

21967350, 2024, 9, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/admi.202300815, Wiley Online Library on [03/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Surface potential distribution studies were carried out using a NtegraAURA scanning probe microscope (NT-MDT) under atmospheric conditions. The surface potential was recorded by the Kelvin probe force microscopy with amplitude modulation (AM-KPFM).

The NiO film morphology was studied using scanning electron microscope (SEM) Supra 25 Zeiss and atomic force microscopy Nano-DST atomic force microscope (Pacific Nanotechnology). To reduce the effect of the surface charge, NiO films were deposited on silicon substrates in one process with the glass substrates for SEM studies. A silicon tip NSG01 from TipsNano with a radius of 10 nm was used. The stiffness coefficient was 1.45–15.1 N m^{-1}. The Gwyddion software package was used for image processing.

The structural perfection of the films was assessed using X-ray diffraction analysis (XRD). Structural and phase analysis of the studied structures was carried out at room temperature on a D8 powder diffractometer (Bruker, Germany) with a vertical Θ-goniometer, in the Bragg–Brentano geometry, radiation generated by X-ray tube with Cu Kα-radiation (λ = 1.5418 Å, ≈8 keV), equipped with a 2D LynxEye CCD detector.

The diagnostics of the atomic composition of the NiO films was carried out by X-ray photoelectron spectroscopy (XPS). Photoemission studies were carried out under ultrahigh vacuum conditions of ≈10^{-9} Torr at room temperature on an Escalab 250Xi complex photoelectron spectrometer (Thermo Fisher Scientific Inc.). System of electron-ion charge compensation was used when recording the photoelectron spectra. Immediately before the experiment, the NiO samples were subjected to heat treatment at a temperature of 120 °C. The energy of the exciting photons was 1486.6 eV, and the spectra of the core levels were measured at an analyzer transmission energy of 20 eV. The standard C 1*s* carbon peak at a binding energy of 284.8 eV was observed in survey spectra for all NiO films and was used to additional calibrate the Ni 2*p* core level spectra. The XPS spectra were decomposed using the Gaussian/Lorentzian peak fitting method. The background for all XPS spectra was subtracted using the smart method of the Thermo Avantage program.

Reflection electron energy loss spectroscopy (REELS) spectra were recorded on an Escalab 250Xi instrument. All NiO samples were irradiated with a flow of monochromatic primary electrons, which were accelerated by a voltage of 1000 V. Recording was carried out in the constant transmission energy mode.

Metal contacts were deposited for conductivity measurements by magnetron sputtering onto NiO films after thermal annealing using lift-off lithography. To form an ohmic contact, Ni was chosen as a contact pair based on the literature data.[21] Contact resistance and resistivity were determined by the transmission line model (TLM) using an Agilent B1500A semiconductor device analyzer. The resistivity values were validated by measurements based on the Hall Effect. The concentration and mobility of the majority charge carriers were determined by the four-probe modified van der Pauw method on an Ecopia HMS 3000 setup.

Optical transmission and reflectance spectra were measured using an AvaSpec-ULS2048XL-EVO-RS spectrometer and a xenon radiation source.

2.3. Statistical Analysis

NiO films were deposited on four glasses, on two glasses in each of the SP1 and SP2 modes. After measuring the NiO thickness, each glass (SP1 and SP2) was divided into fragments. To reduce the scatter in the parameters of the films due to the instability of technological parameters, two samples of NiO films (SP1 and SP2) were annealed in the atmosphere and two in vacuum.

Each type of measurement was performed in two areas on two samples of NiO film obtained under a specific combination of processing conditions. Two series of film measurements (SP1 and SP2) were performed under repeatability conditions: 1) after annealing in atmosphere, 2) after annealing in vacuum. It is worth noting that the studies of samples (SP1 and SP2) after annealing in atmosphere and after annealing in vacuum were carried out on different days, that is, under intermediate reproducibility conditions with one variable parameter "time". To control the measurement reproducibility, the parameters of the NiO films were measured twice.

In this study, the main parameters of the material were transparency, electrical conductivity, as well as bandgap, and work function.

Differences in sample parameters after various treatments could be associated both with a scatter of values due to heterogeneity in the thickness of the sample, caused by the instability of technological process parameters, and with measurement error, depending on the accuracy of the measurement method used. When choosing criteria for assessing the significance of differences in the obtained results, the standard deviation of the results of parameter measurements was used under repeatability conditions or intermediate reproducibility conditions and their heterogeneity, primarily connected with the heterogeneity of the thickness and composition of the NiO film. Keeping in mind the purpose of the study, the average values of two groups of films obtained in different modes were compared: The parameter values of NiO films (SP1 and SP2) were compared after annealing in vacuum and in atmosphere at different temperatures. Keeping in mind the purpose of the study, the average values of the parameters of two groups of films obtained in different modes were compared: NiO (SP1 and SP2) after annealing in vacuum and in atmosphere at different temperatures.

Analysis of XRD spectra was carried out taking into account the Caglioti initial parameters were U = 0.019094, V = −0.024426, and W = 0.013343. Refinement was done using the Thompson– Cox–Hastings (TCH) pseudo-Voigt Axial divergence asymmetry function. The used experimental conditions were $2\theta = 10°–80°$ with a step of 0.02°. For the Al_2O_3 refinement, the TCH profile was used to obtain the instrumental parameters of the equipment, which were added to the instrumental resolution file (IRF) and used later to determine the average crystallite sizes for NiO film.

To accumulate signal intensity to increase measurement sensitivity, XRD, XPS, and UV–vis spectra were measured multiple times in the same area of the sample.

© 2024 The Authors. Advanced Materials Interfaces published by Wiley-VCH GmbH

21967350, 2024, 9, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/admi.202300815, Wiley Online Library on [03/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

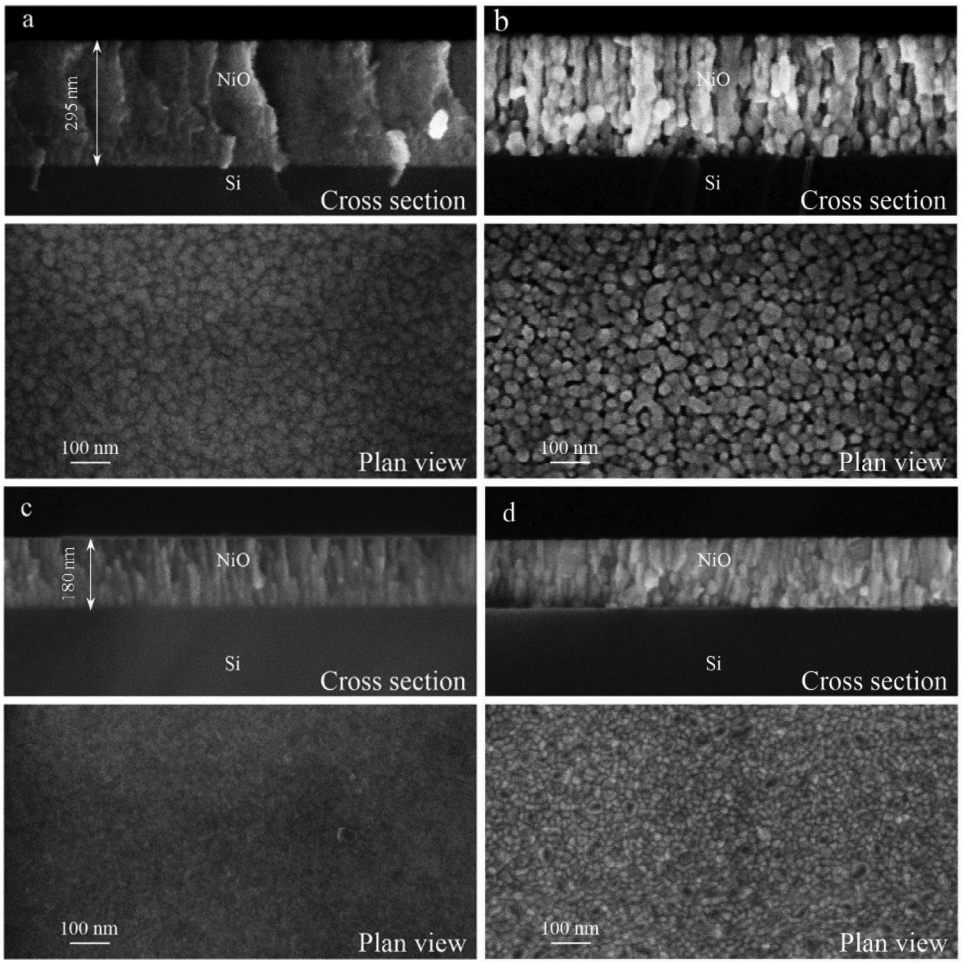


Figure 1. SEM images of the surface and cleavage of a NiO film on a silicon substrate: a) for the film deposited in the SP1 mode and b) subsequent annealing in atmospheric conditions at a temperature of 550 °C, c) for the film deposited in the SP2 mode, and d) subsequent annealing under atmospheric conditions at a temperature of 550 °C.

3. Results and Discussion

3.1. Morphology

The measured NiO film thicknesses for the SP1 and SP2 deposition modes were 146 ± 3 and 179 ± 9 nm, respectively.

SEM images of the surface of the films and their cleavages for two deposition modes and the limiting annealing temperature under atmospheric conditions are shown in **Figure 1**. After deposition in both modes, the films look like a dense homogeneous structure (Figure 1a,c). However, after annealing, the films obtained at low sputtering power (SP1) show a distinct texture of the formation of vertically directed columnar structures (Figure 1b). The average grain diameter on the film surface is 32 nm, which roughly corresponds to the surface texture observed on samples without annealing. At the same time, such a modification of the film structure is not observed on films obtained at high sputtering power (SP2) (Figure 1d). For these films, a clear texture is observed on their surface, but there is no pronounced texture in the cleavage, and the film has a homogeneous morphology. Such a difference in film modification may be due to the formation of an uncompacted amorphous film and subsequent recrystallization for the first deposition mode. It can be assumed that, in this case, the separation of columnar structures plays a predominant role in reducing the conductivity of these films.

The results of AFM on the surface of NiO films also show the presence of grains on their surface. However, it should be noted that the root mean square (RMS) roughness value for films is not large and has a small change during thermal annealing (Figure S1, Supporting Information). For all films immediately after deposition, the RMS values do not exceed 0.8 nm. As the annealing temperature increases to 300 °C, a slight decrease in roughness is observed. However, for all films, a subsequent increase in the annealing temperature leads to a slight increase in roughness to a value of ≈1.2 nm. This also agrees well with the results of SEM studies (increase in film texture).

3.2. Structural Parameters

The results of XRD analysis show the presence of the NiO crystalline phase in the deposited films. In this case, the nature of

© 2024 The Authors. Advanced Materials Interfaces published by Wiley-VCH GmbH

21967350, 2024, 9, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/admi.202300815, Wiley Online Library on [03/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

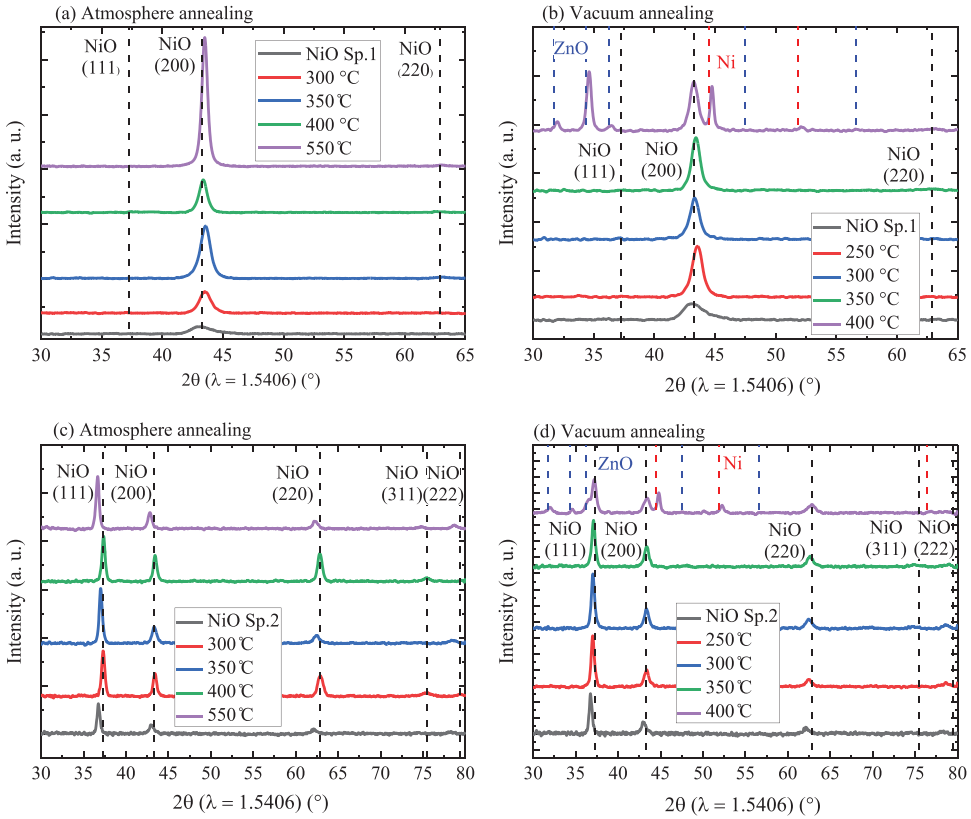


Figure 2. XRD patterns of NiO thin films: a) for SP1 mode and atmosphere annealing, b) for SP1 mode and vacuum annealing, c) for SP2 mode and atmosphere annealing, d) for SP2 mode and vacuum annealing.

the diffraction pattern is very different for films obtained in the two modes described above. **Figure 2** shows XRD patterns of thin NiO films for magnetron sputtering modes with low (a,b) and high (c,d) discharge power with annealing under various conditions. For comparison, the position of theoretical diffraction peaks was obtained using the VESTA program and experimental data[27] corresponding to cubic nickel oxide with the space group $Fm\bar{3}m$. The lattice cell parameters were evaluated using the Rietveld refinement method[28] implemented in the FullProf (Winplotr) software package and further approximation of the experimental data with the average crystallite size evaluation was carried out using Match! – Phase Analysis using Powder Diffraction software. Characteristic XRD pattern approximation is presented in Figure S2a (Supporting Information).

Deviant behavior of the reflex intensity in the XRD patterns and dramatical difference from the simulated XRD pattern, that demonstrated in Figure S2b (Supporting Information), is the evidence of a difference in the preferred film growth orientation in the synthesized samples. Comparing experimental data and simulated profile of XRD patterns clearly seen existence of the texture in the different crystallographic directions. Across the texture direction the crystallite size bigger than other direction, which practically completely agrees with the SEM image analysis.

For films obtained at low discharge power (SP1), one high-intensity peak was observed, corresponding to 43.32°. An increase in the diffraction intensity was observed during film annealing both in an oxygen-containing environment and in vacuum. The results are in good agreement with similar studies.[26] However, in the case of annealing in vacuum at a temperature of 400 °C, the films also exhibited peaks that can be attributed to the cubic Ni phase with the space group $Im\bar{3}m$ and hexagonal ZnO with the space group $P6_3/mc$. The observation of ZnO may be due to diffusion and oxidation of Zn from Corning 0211 glass.

At the same time, for films obtained at high discharge power (SP2) peaks were observed at 2θ angles of 37.32°, 43.32°, 62.92°, 75.4°, and 79.41° corresponding to (111), (200), (220), (311), and (222) cubic lattice plane, respectively. It can be assumed that an increase in the energy of Ni and O atoms during film growth led to the formation of a more densely packed structure. In this case, there is no obvious increase in the intensity of X-ray diffraction after annealing. In addition, as for the first mode of magnetron deposition, in the case of vacuum annealing at a temperature of 400 °C, the formation of Ni and ZnO is also observed. Thus, it is the annealing conditions that affect their formation.

Estimation of the average values of crystallite size (D), Lattice parameter (a), and dislocation density (δ) is given in **Table 1**. The dislocation density (δ) has been evaluated from Williamson–Smallman's formula. It can be seen that the films obtained at a low discharge power had a low crystallinity and a high dislocation density. With an increase in the annealing temperature, a more than twofold increase in the crystallite size and a multiple decrease in the dislocation density are observed. In this case, annealing in a vacuum environment makes it possible to lower the annealing temperature in order to obtain a result corresponding

© 2024 The Authors. Advanced Materials Interfaces published by Wiley-VCH GmbH

21967350, 2024, 9, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/admi.202300815, Wiley Online Library on [03/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 1. XRD results.

Annealing temperature [°C]	Orientation	Crystallite size D[a)], [nm]	Dislocation density δ[b)], [10^{15} m^{-2}]	Lattice parameter a[a)], [Å]
Atmosphere annealing SP1				
-	Texture (200)	4.1	59.5	4.180
	Average	-	-	
300	Texture (200)	7.9	16.0	4.152
	Average	3.0	111.1	
350	Texture (200)	9.7	10.7	4.159
	Average	4.0	62.5	
400	Texture (200)	11.9	7.1	4.169
	Average	8.8	12.9	
550	Texture (200)	19.7	2.6	4.186
	Average	13.4	8.1	
Vacuum annealing SP1				
250	Texture (200)	8.6	13.5	4.210
	Average	9.5	11.1	
300	Texture (200)	10.3	9.4	4.166
	Average	8.5	13.8	
350	Texture (200)	11.7	7.3	4.226
	Average	6.5	23.7	
400	Texture (200)	11.4	7.7	4.189
	Average	7.1	19.8	
Atmosphere annealing SP2				
-	Texture (111)	22.9	1.9	4.220
	Average	14.5	4.8	
300	Texture (111)	21.9	2.1	4.173
	Average	14.7	4.6	
350	Texture (111)	25.5	1.5	4.198
	Average	10.8	8.6	
400	Texture (111)	25.6	1.5	4.171
	Average	17.3	3.3	
550	Texture (111)	24.9	1.6	4.224
	Average	16.6	3.6	
Vacuum annealing SP2				
250	Texture (111)	27.6	1.3	4.193
	Average	10.2	9.6	
300	Texture (111)	21.7	2.1	4.190
	Average	10.7	8.7	
350	Texture (111)	20.0	2.5	4.191
	Average	10.2	9.6	
400	Texture (111)	17.9	3.1	4.188
	Average	8.9	12.6	

[a)] 1% (repeatability precision), 2% (intermediate precision); [b)] 2% (repeatability precision), 4% (intermediate precision).

© 2024 The Authors. Advanced Materials Interfaces published by Wiley-VCH GmbH

21967350, 2024, 9, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/admi.202300815, Wiley Online Library on [03/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

to atmospheric annealing. At the same time, films obtained at high sputtering power show atypical behavior. These films show a much higher crystallite size directly after film deposition. Subsequent annealing leads to an insignificant change in crystallinity and to a decrease in the dislocation density. This trend is observed for samples annealed in an oxygen-containing environment. In the case of annealing in vacuum, for samples obtained at high power, an inverse relation is observed, where the crystallite size decreases by more than 10%. It can be assumed that these conditions favor the formation of oxygen vacancies in the NiO films.

3.3. XPS Studies

XPS measurements were carried out to study the chemical composition and degree of oxidation of nickel oxide films. **Figure 3** shows the XPS spectra of the Ni $2p_{3/2}$ core level, recorded for NiO films formed under different modes of deposition and subsequent annealing. All observed XPS spectra have a developed structure with distinct peaks. The presence of residual contaminants, such as hydroxides, is possible on the NiO surface. This may lead to incorrect interpretations of the Ni 2*p* spectrum for quantification. Therefore, we make only qualitative comparison of peak area ratios. The spectra confirm the presence of the main oxygen-containing compounds NiO and Ni_2O_3 in the films. Peaks at binding energies of 853.8 and 855.6 eV represent different nickel states Ni^{2+} and Ni^{3+}, respectively. Peaks at binding energies of 860.8, 864.0, and 866.4 eV are satellite and are due to shake-up processes. The position and shape of the Ni 2*p* core level coincide with the literature data.[29–31] The Ni^{2+} peak corresponds to the Ni—O bond. The Ni^{3+} is due to metal deficiency. Ni^{3+} ions induced by quasi-localized holes around Ni^{2+} vacancies in the lattice generate p-type conductivity in the NiO thin film.[32] A thorough analysis of the decomposed XPS spectra showed that the ratio of the Ni^{3+}/Ni^{2+} peak areas decreased with increasing annealing temperature for the samples obtained in the SP1 mode with annealing in an oxygen-containing environment and the SP1 and SP2 modes with annealing in vacuum conditions (see Figure 3a,b,d). This is due to a change in the oxidation state of nickel, the conversion of Ni^{3+} ions into Ni^{2+} (transition from Ni_2O_3 to NiO), and, accordingly, a decrease in point defects. Similar results were obtained in work.[33] However, for samples prepared in the SP2 mode and annealed in an oxygen-containing environment, the Ni^{3+}/Ni^{2+} peak area ratio increases with the annealing temperature (see Figure 3c). Thus, there are more Ni^{3+} ions than Ni^{2+}, and the number of defects increases. The conditions of film deposition at a high discharge power in an oxygen-containing environment favor the formation of oxygen vacancies in NiO films. The results of XPS studies are in good agreement with studies of XRD and film conductivity.

A distinctive feature of the spectra for samples deposited in the SP2 mode and annealed in an oxygen-containing environment at low annealing temperatures is the presence of a peak at a binding energy of 852.5 eV, which can definitely be attributed to the presence of Ni^0 (see Figure 3c). For NiO films prepared in other modes, this peak is not observed. This means that in the SP2 mode samples, during annealing in an oxygen-containing environment, nickel partially materializes in the metallic form. With an increase in the annealing temperature, nickel is oxidized predominantly to the Ni^{3+} state, and already at an annealing temperature of 400 °C, the Ni^0 peak disappears. Thus, the relative concentration of nickel with different oxidation states can be controlled by choosing the appropriate film deposition mode.

It should be noted that for the samples obtained in the SP1 and SP2 modes during vacuum annealing, already at an annealing temperature of 300 °C, a peak appeared in the spectrum of the Ni 2*p* core level at a binding energy of 852.6 eV. It can be assumed that this peak is associated with nickel reduction. Since the NiO films were grown on glass containing zinc, under the influence of temperature during annealing NiO, Zn penetrates into the volume of the sample and diffuses to the surface, where it interacts with O, which is reflected in the appearance and increase of the peak intensity at lower binding energies from Ni^{2+}. The presence of the Zn 2*p* and Zn 3*p* core levels was also found in the survey spectra for these samples at the corresponding annealing temperatures. This result is in good agreement with the results of XRD studies, where peaks corresponding to Ni and ZnO are observed in the diffraction patterns for the samples obtained in SP1 and SP2 modes after vacuum annealing at a temperature of 400 °C (see Figure 2b,d).

Figure S3 (Supporting Information) presents XPS spectra of the O 1*s* core level recorded for NiO films under various deposition and subsequent annealing conditions. The main peak of the O 1*s* core level, located at ≈529.1 eV, is designated Ni^{2+} and is attributed to the Ni—O bond. The peak at ≈531.4 eV is attributed to the formation of a Ni vacancy. A small peak at ≈533.0 eV may be due to adsorbed water or possibly adsorbed O_2. The trend in the O 1*s* core level spectra also correlates well with measurements of the electrical conductivity of the samples.

3.4. Electrical Parameters

At the initial stage, the dynamics of changes in the resistivity of NiO films with the annealing time for various annealing temperatures in an oxygen-containing environment was determined. In this case, an ohmic contact with a linear current–voltage characteristic was observed on all samples. The dependence of the change in the resistivity of the NiO film deposited in the SP1 mode with annealing in an oxygen-containing environment on the annealing time is shown in **Figure 4a**.

After deposition in the SP1 mode, the NiO films demonstrated high conductivity, and the resistivity was 0.01 Ω cm. Annealing in an atmospheric environment showed a significant decrease in the film conductivity at temperatures ≈300 °C. This may indicate the relaxation of intrinsic defects, namely, V_{Ni}. It should be noted that at an annealing duration of >60 min, the change in conductivity is not significant, which indicates the stabilization of the process.

The results of measuring the resistivity by the TLM method and the Hall method for various annealing conditions for 60 min are shown in Figure 4b. It is worth noting that due to the high resistivity on some samples, Hall measurement was not possible. It can be seen from the dependence that the samples formed at a low deposition power (SP1) demonstrate a sharp increase in resistivity even at an annealing temperature of 250 °C. At the same time, heat treatment of NiO films in a vacuum environment also leads to a decrease in the electrical conductivity of the

© 2024 The Authors. Advanced Materials Interfaces published by Wiley-VCH GmbH

21967350, 2024, 9, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/admi.202300815, Wiley Online Library on [03/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

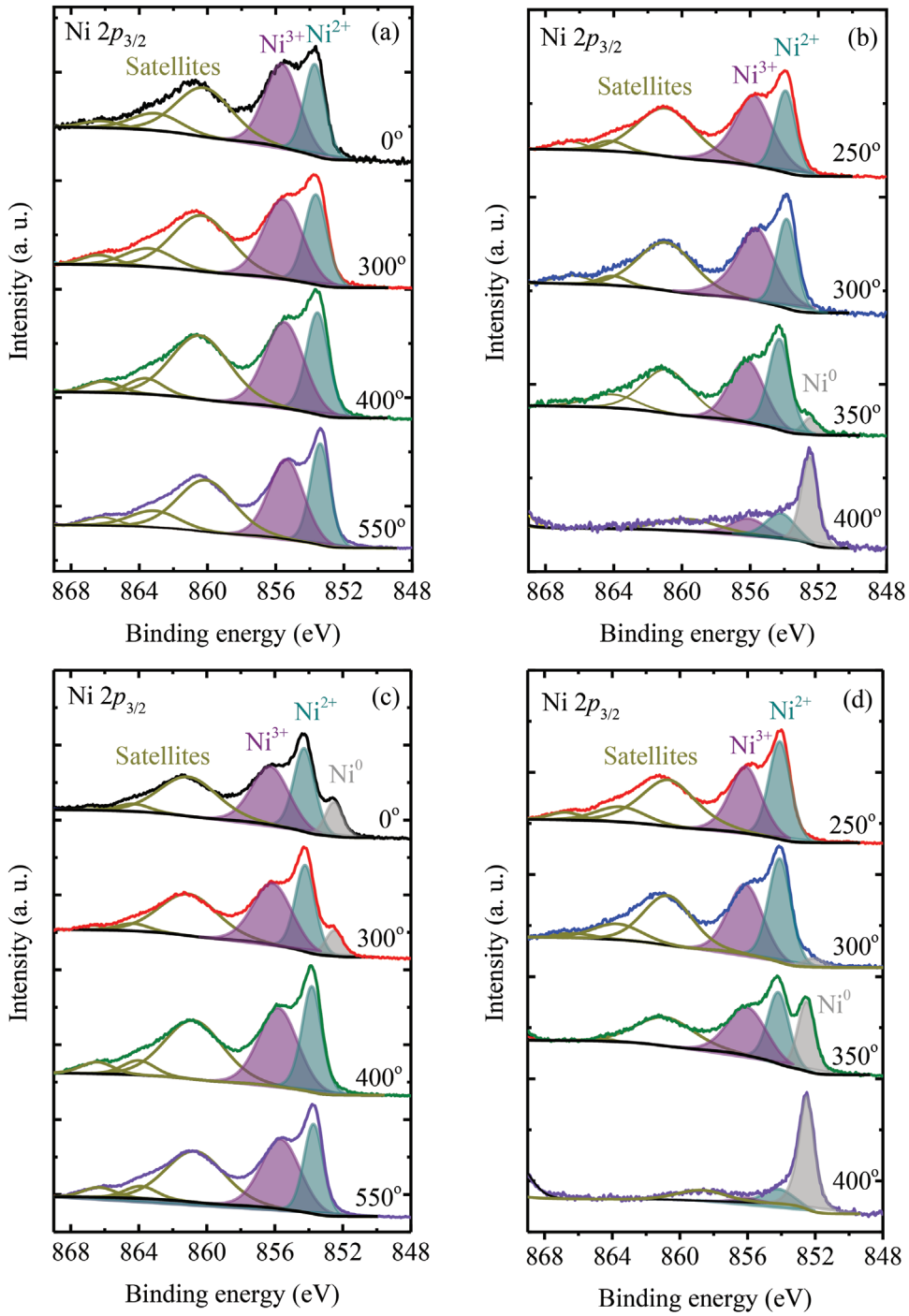


Figure 3. Ni $2p_{3/2}$ core level spectra of NiO films. a) for SP1 mode and atmosphere annealing, b) for SP1 mode and vacuum annealing, c) for SP2 mode and atmosphere annealing, d) for SP2 mode and vacuum annealing.

layers (Figure 4b). This behavior is typical for the annealing of NiO films obtained by magnetron sputtering.[15,22,23] In this case, in contrast to annealing in an oxygen-containing atmosphere, the changes in conductivity are more pronounced.

At the same time, films obtained at a high sputtering power (in the SP2 mode) show a different behavior. Initially, after deposition, the films had a high resistivity. An increase in the annealing temperature to 300 °C, as in the first case, led to a further increase in the resistance of the films. At the same time, annealing at temperatures ≈350 °C led to a significant decrease in resistivity. It is worth noting that this behavior was not recorded for samples annealed in vacuum. It can be assumed that the presence of oxygen is of primary importance for these films. However, it was impossible to determine the further dynamics of changes in resistivity

© 2024 The Authors. Advanced Materials Interfaces published by Wiley-VCH GmbH

21967350, 2024, 9, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/admi.202300815, Wiley Online Library on [03/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

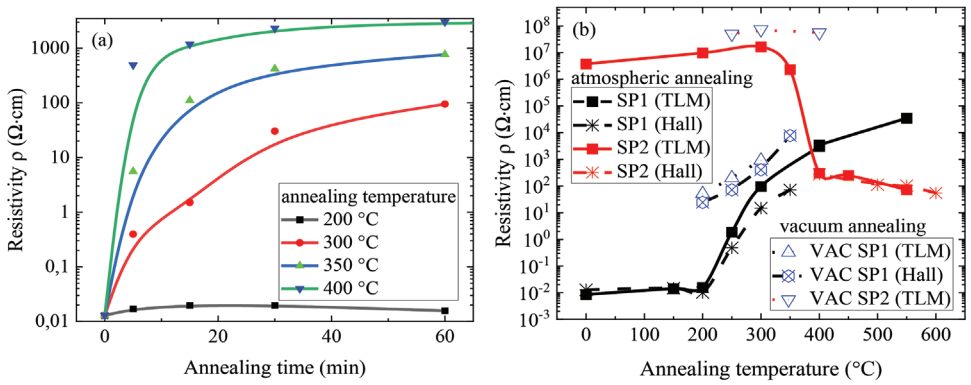


Figure 4. a) Dependence of resistivity (deposition mode SP1) on annealing time for various heat treatment conditions; b) Dependences of resistivity on annealing temperature for various film deposition modes (SP1 and SP2) and for annealing in atmosphere and vacuum.

for annealing in a vacuum environment due to the formation of additional glass impurities in the NiO film. The obtained values correlate with the change in the crystallinity of the film structure and the dislocation density.

The dependences of the concentration and mobility of charge carriers of NiO films are presented in **Figure 5**. All films for which it was possible to conduct studies using the Hall Effect had *p*-type conductivity. For films deposited at low sputtering power, annealing at low temperatures leads to a slight increase in the concentration of charge carriers. At annealing temperatures of 250–350 °C, the concentration of charge carriers decreases sharply. A further increase in the annealing temperature does not lead to any significant changes.

3.5. Optical Characteristics

After deposition in the SP1 mode of the NiO film, the layers had low transparency and a terracotta tint. At the same time, the samples obtained in the SP2 mode had high transparency and a yellow tint. Heat treatment of the samples led to an increase in their transparency. The dependence of the transmittance of the NiO samples obtained in the low power mode on the annealing time under atmospheric conditions is shown in **Figure 6**. Similar to the dynamics of changes in resistivity, the change in transparency is sharp at an annealing time of 30 min. Continued annealing also leads to an increase in transparency, but its character is no longer so sharp.

The experimental spectral characteristics of transmission, reflection, and absorption of NiO samples obtained in the SP1 mode and annealed under atmospheric conditions at various temperatures and annealing times are presented in **Figure 7a**. The spectral characteristics show a trend toward an increase in the transparency of the samples and a decrease in their absorption in the visible region depending on the heat treatment temperature. It can be seen that this trend begins at temperature of 300 °C, which correlates with the change in electrical conductivity (Figure 4a).

In the case of heat treatment in vacuum (Figure 7b), there is also a tendency to increase the transparency of the films. At the same time, higher transparency is achieved at lower temperatures. However, it can be seen that in comparison with heat treatment under atmospheric conditions, a shift in the spectral characteristics to the long wavelength region is observed.

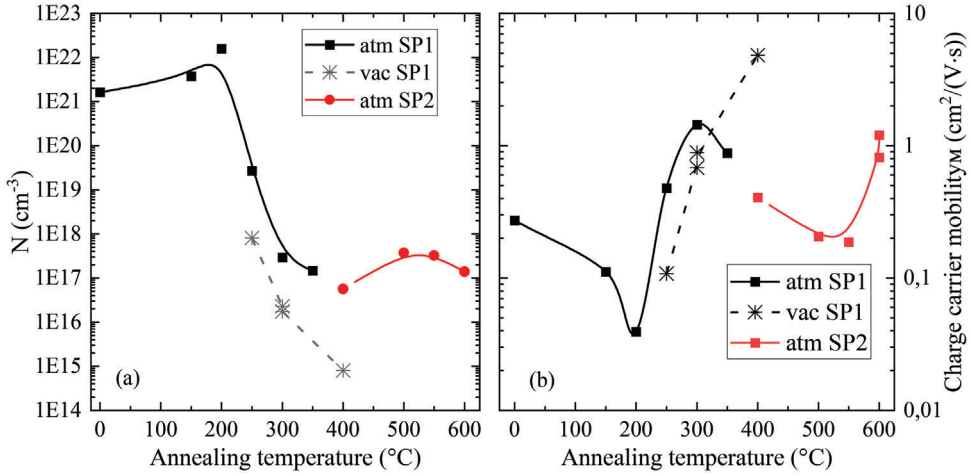


Figure 5. Dependences of the concentration a) and mobility of charge carriers b) on the annealing temperature for various film deposition modes (SP1 and SP2) and for annealing in the atmosphere and vacuum.

© 2024 The Authors. Advanced Materials Interfaces published by Wiley-VCH GmbH

21967350, 2024, 9, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/admi.202300815, Wiley Online Library on [03/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

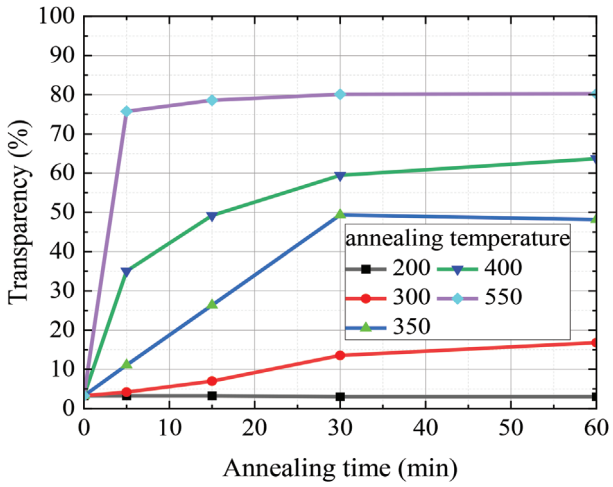


Figure 6. Dependence of the transmittance on the annealing time of the samples obtained in the SP1 mode with annealing under atmospheric conditions at different temperatures (λ = 700 nm)

Spectral characteristics for samples obtained in SP2 mode are shown in Figure 7c. As for the samples obtained in the SP1 mode, there is a tendency to increase the film transparency with an increase in the heat treatment temperature, but the increase in transparency is not so pronounced. In addition, a more pronounced shift of the spectral characteristics to the long wavelength region is also observed than for the samples obtained in the SP1 mode.

It should be noted that after heat treatment in vacuum at a temperature of 400 °C (Figure 7d), NiO samples have low transparency, and a change in the shade of the front surface of the film occurs. This is clearly seen in the spectral characteristics (Figure 7). With a decrease in transparency, there is an increase in absorption in the visible region, even compared to the initial film.

The optical bandgap was estimated using Tauc plot. The absorption coefficient (α) of the NiO film is obtained using the following equation[34]

$$\alpha = \frac{1}{d} ln\left(\frac{(1-R)^2}{T}\right) \quad (1)$$

where d, R, and T are the thickness, reflectance, and transmittance of the NiO film, respectively. Further, the relation between absorption coefficient (α) and incident photon energy ($h\nu$) can be written as

$$\alpha h\nu = A\left(h\nu - E_g\right)^n \quad (2)$$

where A is an energy-independent constant; E_g is optical energy bandgap of the material and the exponent n depends on the type of transition. For crystalline semiconductors, n = 1/2, 2 values corresponding to the allowed direct, and allowed indirect transitions, respectively.

Figure 8 shows the Tauc Plot for various annealing modes and temperatures. It can be seen that for the films obtained at a low deposition power (SP1), radiation absorption is observed in the sub-band bandgap energy. This can be caused by both a high density of film defects and a high density of oxygen vacancies responsible for the high conductivity of films.[35] As in the method described by Tauc, the linear fit of the fundamental peak is applied. Additionally, a linear fit used as an abscissa is applied for the slope below the fundamental absorption. An intersection of the two fitting lines gives the bandgap energy estimation.[35] The presence of intraband absorption was also noticed on samples that underwent heat treatment in a vacuum environment at temperatures of 400 °C. Thus, for these samples, the described method was also used to determine the optical bandgap. In the case of NiO films obtained at high sputtering power (SP2), the dependences do not show pronounced intraband transitions, which indicates the absence of interband levels.

The values of the optical bandgap versus the annealing temperature for various deposition modes and annealing conditions are presented in **Table 2**. In the case of NiO deposition at low power, a rather high value of the optical bandgap is observed. Subsequent annealing leads to a sharp decrease in the value of E_g. However, an increase in the annealing temperature is accompanied by its growth. In this case, this increase has a sharper character for annealing under vacuum conditions. At the same time, films obtained at high sputtering power have an initially low bandgap. As in the first case, an increase in E_g is observed with an increase in the annealing temperature. We note that, as in the first case, annealing in vacuum leads to a more intense increase in E_g.

Using the REELS method, the bandgap of the NiO films was also determined. The results are presented in Table 2. The result is different from the values obtained from the optical measurements. However, the discrepancy between the values of the bandgap is also observed in other studies, for example.[36] For NiO samples obtained in the SP2 mode under vacuum annealing conditions, it was not possible to determine the bandgap from the REELS spectra.

Using the AM-KPFM method, the surface potential was recorded. The work function of the studied samples was determined by comparing the potential of the sample V_{sam} and the potential of the test calibration sample V_{test} with the known work function A_{test} (highly oriented pyrolytic graphite). The work function was determined by the formula $A_{sam} = A_{test}+e(V_{test}-V_{sam})$. The change in the measured bandgap by the REELS method correlates well with the change in the measured work function by the AM-KPFM method (see Table 2).

In general, the change in the optical characteristics, as well as the electrical characteristics, can be explained by the relaxation of the film's intrinsic defects and a decrease in subband absorption. In addition, the shift in the spectral characteristics corresponds to the visual appearance of the samples.

4. Conclusion

In summary, NiO films were deposited on glass substrates by dc magnetron sputtering. The effect of the discharge power and annealing modes of NiO films on their physical properties has been studied. The XRD data show that the crystallographic orientation strongly depends on the discharge power during the deposition of NiO thin films. The XRD results demonstrate a polycrystalline nature with an fcc phase with a preferred orientation of (200) for the low discharge power mode and (111) for the high discharge power mode. An increase in the annealing temperature leads to

© 2024 The Authors. Advanced Materials Interfaces published by Wiley-VCH GmbH

21967350, 2024, 9, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/admi.202300815, Wiley Online Library on [03/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

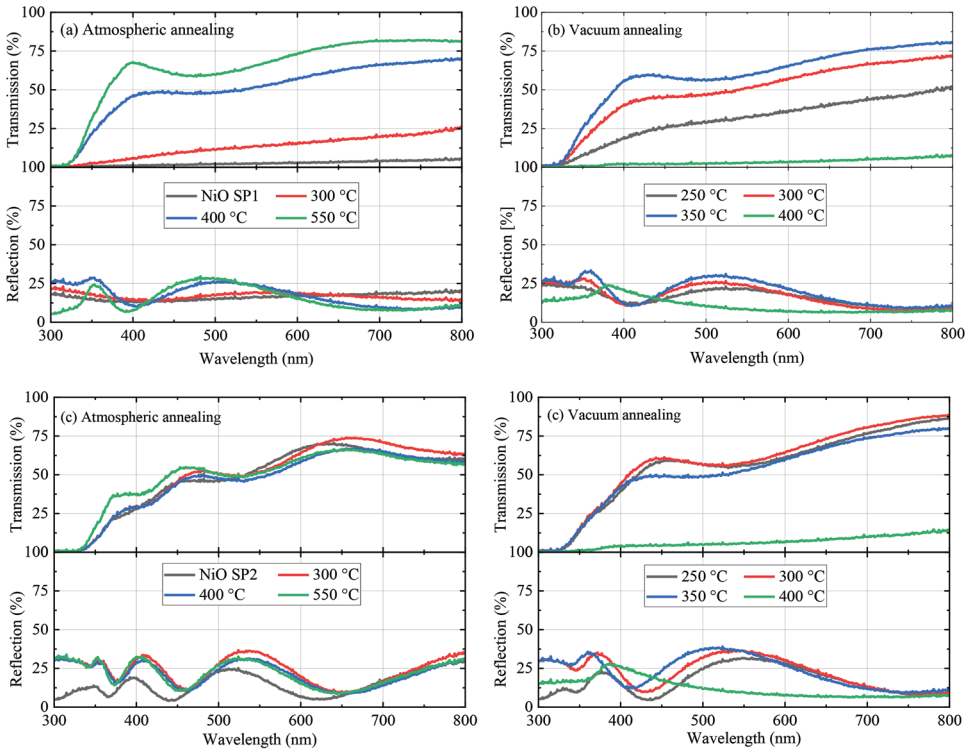


Figure 7. Transmission and reflection spectra of NiO samples obtained in the SP1 a,b) and SP2 c,d) modes with annealing under atmospheric and vacuum conditions at various annealing temperatures.

a decrease in the dislocation density regardless of the NiO film deposition mode. Optical measurements indicate an increase in the transparency of the films and a decrease in their absorption in the visible region with an increase in the annealing temperature both under atmospheric and vacuum conditions; the calculated value of the bandgap varied from 3.50 to 3.74 eV. AFM analysis showed that the surface roughness of the films is practically independent of the deposition mode and annealing conditions. The XPS decomposition of the Ni 2*p* core level spectra showed that the relative atomic percentages varied systematically with the annealing temperature of the NiO films. An increase in the annealing temperature during deposition in the low-power mode during annealing in an oxygen-containing environment and in the modes of low and high-power during annealing in vacuum leads

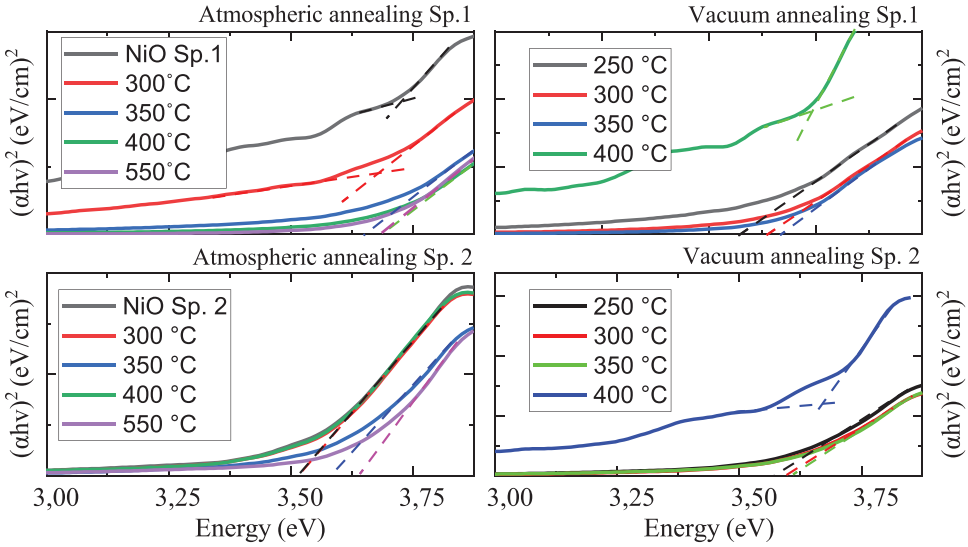


Figure 8. Tauc Plot of $(\alpha h\nu)^2$ versus $h\nu$ of NiO thin films deposited at different sputtering power and post-deposition annealing temperature.

© 2024 The Authors. Advanced Materials Interfaces published by Wiley-VCH GmbH

21967350, 2024, 9, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/admi.202300815, Wiley Online Library on [03/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 2. Bandgap and work function values.

Annealing temperature [°C]	Optical bandgap, [eV]		Bandgap (REELS), [eV]		Work function (AM-KPFM), [eV]	
	SP1	SP2	SP1	SP2	SP1	SP2
Atmosphere						
0	3.73	3.50	3.55	3.60	4.95 eV	4.85 eV
300	3.68	3.51	3.45	3.50	4.85 eV	4.80 eV
400	3.68	3.58	3.45	3.50	4.90 eV	4.90 eV
550	3.69	3.63	3.60	3.55	5.00 eV	5.00 eV
Vacuum						
250	3.57	3.57	3.60	–	4.70 eV	5.05 eV
300	3.63	3.59	3.75	–	4.80 eV	5.00 eV
350	3.66	3.60	3.85	–	5.00 eV	4.85 eV
400	3.74	3.68	3.55	–	4.65 eV	4.90 eV

to the formation of Ni^{2+} ions and, accordingly, to a decrease in conductivity. However, an increase in the annealing temperature during deposition in the high-power mode during annealing in an oxygen-containing environment leads to the formation of Ni^{3+} ions and an increase in conductivity. It should be noted that for the samples obtained in the SP1 and SP2 modes with vacuum annealing at temperatures ≈300 °C, Zn penetrates from the glass substrate into the NiO film and diffuses to the surface, where ZnO is formed, which is confirmed by XRD and XPS studies.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

The study was carried out at an Alferov University with the support by the Russian Science Foundation Grant No. 22-29-20181, https://rscf.ru/en/project/22-29-20181/ (https://rscf.ru/project/22-29-20181/), and Ministry of Science and Higher Education of the Russian Federation (Agreement No. 075-15-2022-1150). XPS studies were performed using equipment from the Resource Center "Physical Methods of Surface Investigation" of the Scientific Park of St. Petersburg State University. Vjaceslavs Bobrovs acknowledges the support of the Latvian Council of Science, project: DNSSN, № lzp-2021/1-0048

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

DC magnetron sputtering, electrical conductivity, NiO thin films, oxides, X-ray photoelectron spectroscopy

Received: September 27, 2023
Revised: December 29, 2023
Published online: January 12, 2024

[1] B. V. Mistry, P. Bhatt, K. H. Bhavsar, S. J. Trivedi, U. N. Trivedi, U. S. Joshi, *Thin Solid Films* **2011**, *519*, 3840.
[2] H. Zhang, J. Cheng, F. Lin, H. He, J. Mao, K. S. Wong, A. K.-Y. Jen, W. C. H. Choy, *ACS Nano* **2016**, *10*, 1503.
[3] Z. Liu, A. Zhu, F. Cai, L. Tao, Y. Zhou, Z. Zhao, Q. Chen, Y.-B. Cheng, H. Zhou, *J. Mater. Chem. A* **2017**, *5*, 6597.
[4] E. Aydin, J. Troughton, M. D. Bastiani, E. Ugur, M. Sajjad, A. Alzahrani, M. Neophytou, U. Schwingenschlögl, F. Laquai, D. Baran, S. De Wolf, *ACS Appl. Energy Mater.* **2018**, *1*, 6227.
[5] S. Sajid, A. M. Elseman, H. Huang, J. Ji, S. Dou, H. Jiang, X. Liu, D. Wei, P. Cui, M. Li, *Nano Energy* **2018**, *51*, 408.
[6] N. P. Klochko, V. R. Kopach, I. I. Tyukhov, D. O. Zhadan, K. S. Klepikova, G. S. Khrypunov, S. I. Petrushenko, V. M. Lyubov, M. V. Kirichenko, S. V. Dukarov, A. L. Khrypunova, *Sol. Energy* **2018**, *164*, 149.
[7] X. Chang, Y.-F. Wang, X. Zhang, Z. Liu, J. Fu, S. Fan, R. Bu, J. Zhang, W. Wang, H.-X. Wang, J. Wang, *Appl. Phys. Lett.* **2018**, *112*, 032103.
[8] N. Park, K. Sun, Z. Sun, Y. Jing, D. Wang, *J. Mater. Chem. C* **2013**, *1*, 7333.
[9] M. D. Irwin, D. B. Buchholz, A. W. Hains, R. P. H. Chang, T. J. Marks, *Proc. Natl. Acad. Sci. U. S. A.* **2008**, *105*, 2783.
[10] H. Sato, T. Minami, S. Takata, T. Yamada, *Thin Solid Films* **1993**, *236*, 27.
[11] K. O. Egbo, C. P. Liu, C. E. Ekuma, K. M. Yu, *J. Appl. Phys.* **2020**, *128*, 135705.
[12] D. Y. Jiang, J. M. Qin, X. Wang, S. Gao, Q. C. Liang, J. X. Zhao, *Vacuum* **2012**, *86*, 1083.
[13] I. Hotový, D. Búc, S. Haščík, O. Nennewitz, *Vacuum* **1998**, *50*, 41.
[14] S.-J. Huang, C.-W. Chou, Y.-K. Su, J.-H. Lin, H.-C. Yu, D.-L. Chen, J.-L. Ruan, *Appl. Surf. Sci.* **2017**, *401*, 373.
[15] Y. Zhao, H. Wang, F. Yang, Z. Zhen, X. Li, Q. Li, J. Li, *Vacuum* **2018**, *151*, 163.
[16] H. L. Lu, G. Scarel, M. Alia, M. Fanciulli, S.-J. Ding, D. W. Zhang, *Appl. Phys. Lett.* **2008**, *92*, 222907.

© 2024 The Authors. Advanced Materials Interfaces published by Wiley-VCH GmbH

21967350, 2024, 9, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/admi.202300815, Wiley Online Library on [03/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

[17] M. Yang, Z. Shi, J. Feng, H. Pu, G. Li, J. Zhou, Q. Zhang, *Thin Solid Films* **2011**, *519*, 3021.
[18] L. De Los Santos Valladares, A. Ionescu, S. Holmes, C. H. W. Barnes, A. Bustamante Domínguez, O. Avalos Quispe, J. C. González, S. Milana, M. Barbone, A. C. Ferrari, H. Ramos, Y. Majima, *J. Vac. Sci. Technol. B* **2014**, *32*, 051808.
[19] Z. Jia, L. Wang, N. Zhang, T. Ren, J. J. Liou, *Appl. Phys. Lett.* **2013**, *102*, 042901.
[20] H. Wang, G. Wu, X. P. Cai, Y. Zhao, Z. F. Shi, J. Wang, X. C. Xia, X. Dong, B. L. Zhang, Y. Ma, G. T. Du, *Vacuum* **2012**, *86*, 2044.
[21] Y. Ren, L. Li, N. Liu, K. Zhang, C. Li, Z. Chen, B. Zhang, *Vacuum* **2020**, *182*, 109784.
[22] L. Li, X. Wang, Y. Liu, J.-P. Ao, *J. Vac. Sci. Technol. A* **2016**, *34*, 02D104.
[23] R. Karsthof, H. Von Wenckstern, V. S. Olsen, M. Grundmann, *APL Mater.* **2020**, *8*, 121106.
[24] S. Lany, J. Osorio-Guillén, A. Zunger, *Phys. Rev. B* **2007**, *75*, 241203.
[25] S.-K. Kim, H. Yang, Y.-S. Kim, *J. Mater. Sci.* **2020**, *55*, 17046.
[26] M. Wang, Y. Thimont, L. Presmanes, X. Diao, A. Barnabé, *Appl. Surf. Sci.* **2017**, *419*, 795.
[27] A. C. Malingowski, P. W. Stephens, A. Huq, Q. Huang, S. Khalid, P. G. Khalifah, *Inorg. Chem.* **2012**, *51*, 6096.
[28] L. B. Mccusker, R. B. Von Dreele, D. E. Cox, D. Louër, P. Scardi, *J. Appl. Crystallogr.* **1999**, *32*, 36.
[29] M. C. Biesinger, B. P. Payne, L. W. M. Lau, A. Gerson, R. S. C. Smart, *Surf. Interface Anal.* **2009**, *41*, 324.
[30] P. Dubey, N. Kaurav, R. S. Devan, G. S. Okram, Y. K. Kuo, *RSC Adv.* **2018**, *8*, 5882.
[31] M. Imran, H. Coskun, N. A. Khan, J. Ouyang, *Appl. Phys. A* **2021**, *127*, 117.
[32] U. Kwon, B.-G. Kim, D. C. Nguyen, J.-H. Park, N. Y. Ha, S.-J. Kim, S. H. Ko, S. Lee, D. Lee, H. J. Park, *Sci. Rep.* **2016**, *6*, 30759.
[33] H.-L. Chang, T.-R. Jeng, J.-P. Chen, W.-H. Yen, P. Yen, D. Huang, J.-J. Ju, *Jpn. J. Appl. Phys.* **2005**, *44*, 6109.
[34] M. Patel, H.-S. Kim, J. Kim, *Adv. Electron. Mater.* **2015**, *1*, 1500232.
[35] P. Makula, M. Pacia, W. Macyk, *J. Phys. Chem. Lett.* **2018**, *9*, 6814.
[36] N. Chen, D. Wang, J. Hu, L. Guan, Z.-H. Lu, *Phys. Status Solidi B* **2022**, *259*, 2100459.

© 2024 The Authors. Advanced Materials Interfaces published by Wiley-VCH GmbH

21967350, 2024, 9, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/admi.202300815, Wiley Online Library on [03/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License